DATABASE MANAGEMENT SYSTEM AND NORMALIZATION

A Database Management System (DBMS) is software used to create, store, organize, retrieve, and manage data in a database. Examples of DBMS software include MySQL, PostgreSQL, Oracle Database, and Microsoft SQL Server.

A database table consists of rows and columns. A row represents a record, while a column represents an attribute of the data.

A PRIMARY KEY is a column or combination of columns that uniquely identifies each record in a table. A primary key cannot contain duplicate values and cannot contain NULL values.

A FOREIGN KEY is a column or group of columns that creates a relationship between two tables. It usually refers to the primary key of another table. Foreign keys help maintain referential integrity between related tables.

For example, consider a STUDENT table:

Student_ID | Student_Name | Department_ID

Student_ID can be the primary key because each student has a unique Student_ID.

Consider another table:

DEPARTMENT

Department_ID | Department_Name

Department_ID can be the primary key of the DEPARTMENT table. Department_ID in the STUDENT table can be a foreign key referring to the DEPARTMENT table.

DATABASE NORMALIZATION

Normalization is a database design technique used to organize data and reduce data redundancy. It divides large tables into smaller related tables and establishes relationships between them.

The main goals of normalization are to reduce duplicate data, improve data consistency, and prevent data anomalies.

DATA REDUNDANCY

Data redundancy occurs when the same information is stored unnecessarily in multiple places. Redundant data increases storage requirements and can cause inconsistencies when one copy is updated but another copy is not.

UPDATE ANOMALY

An update anomaly occurs when the same piece of information is stored in multiple rows and a change must be made in several places. If some rows are updated and others are not, the database may contain inconsistent information.

INSERTION ANOMALY

An insertion anomaly occurs when it is difficult or impossible to insert new information because another unrelated piece of information is also required.

DELETION ANOMALY

A deletion anomaly occurs when deleting one record unintentionally removes other important information.

FIRST NORMAL FORM

A table is in First Normal Form (1NF) when each column contains atomic values and there are no repeating groups.

For example, a Student table should not store multiple phone numbers in a single column such as:

Phone_Number = "9876543210, 9123456780"

Instead, the values should be stored separately according to the database design.

SECOND NORMAL FORM

A table is in Second Normal Form (2NF) when it is already in 1NF and every non-key attribute is fully dependent on the entire primary key.

2NF mainly deals with partial dependency.

Partial dependency occurs when a non-key attribute depends only on part of a composite primary key rather than the complete key.

THIRD NORMAL FORM

A table is in Third Normal Form (3NF) when it is already in 2NF and there is no transitive dependency of non-key attributes on the primary key.

A transitive dependency occurs when a non-key attribute depends on another non-key attribute instead of directly depending on the primary key.

For example:

Student_ID -> Department_ID -> Department_Name

Here, Department_Name depends on Department_ID, not directly on Student_ID. This can create a transitive dependency.

Normalization improves database design by reducing redundancy and preventing insertion, update, and deletion anomalies.

However, highly normalized databases may require more joins when retrieving data from multiple related tables. Therefore, database designers sometimes use denormalization when faster read performance is more important than reducing redundancy.

ACID PROPERTIES

Database transactions commonly follow ACID properties.

ATOMICITY means that a transaction is treated as a single unit. Either all operations of the transaction are completed or none of them are applied.

CONSISTENCY ensures that a transaction changes the database from one valid state to another valid state while maintaining database rules.

ISOLATION ensures that concurrent transactions do not improperly interfere with each other.

DURABILITY means that once a transaction has been successfully committed, its changes are preserved even if a system failure occurs.

TRANSACTION EXAMPLE

Consider transferring ₹5,000 from Account A to Account B.

The transaction involves two operations:

1. Deduct ₹5,000 from Account A.
2. Add ₹5,000 to Account B.

If the deduction succeeds but the addition fails, the transaction should not leave the database in an incorrect state. Atomicity ensures that the complete transaction succeeds or the changes are rolled back.